WIRELESS TECHNOLOGY

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1. Bluetooth

- **Theory:** Bluetooth is a short-range wireless communication technology primarily used for connecting devices over short distances. It operates in the 2.4 GHz ISM band and is used in a wide variety of devices like headphones, speakers, and keyboards. Bluetooth is known for its low power consumption, which makes it ideal for battery-powered devices. It supports both point-to-point and point-to-multipoint connections, allowing for flexible communication between devices.

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2. Wi-Fi

- **Theory:** Wi-Fi is a wireless networking technology that allows devices to communicate over a local area network (LAN). It uses the IEEE 802.11 standard and operates in the 2.4 GHz and 5 GHz frequency bands. Wi-Fi is widely used for high-speed internet access, streaming, and connecting devices within a home or business environment. Modern Wi-Fi standards like Wi-Fi 5 (802.11ac) and Wi-Fi 6 (802.11ax) provide high data transfer rates and improved efficiency.

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3. Zigbee

- **Theory:** Zigbee is a wireless communication protocol used primarily for low-power, low-data-rate applications. It operates in the 2.4 GHz frequency band and is often used in home automation, smart homes, and industrial applications. Zigbee is known for its mesh networking capability, which allows devices to communicate with each other directly or through intermediate nodes, extending its range.

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4. LoRa (Long Range)

- **Theory:** LoRa is a low-power, long-range wireless communication technology designed for the Internet of Things (IoT). It operates in the sub-GHz frequency bands and is widely used in applications like smart agriculture, environmental monitoring, and asset tracking. LoRa uses spread spectrum modulation to achieve long communication ranges while consuming very little power.

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5. 5G

- **Theory:** 5G is the fifth generation of mobile communication technology, designed to provide faster speeds, lower latency, and increased capacity compared to previous generations like 4G. 5G networks use higher frequency bands (millimeter waves) and advanced technologies like MIMO (Multiple Input Multiple Output) and beamforming to deliver high-speed data transfer, making it ideal for applications like autonomous vehicles, augmented reality, and industrial automation.

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6. NFC (Near Field Communication)

- **Theory:** NFC is a short-range wireless communication technology used for exchanging data between devices over distances of just a few centimeters. It operates at 13.56 MHz and is commonly used in contactless payment systems, access control, and data exchange between smartphones. NFC is a subset of RFID (Radio Frequency Identification) and allows for quick and secure communication.

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7. Infrared (IR)

- **Theory:** Infrared communication is a wireless technology that uses infrared light to transmit data over short distances. It is typically used in remote controls, short-range data transmission, and simple signaling. Infrared communication requires a direct line of sight between the transmitting and receiving devices, making it less flexible compared to other wireless technologies.

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8. Satellite Communication

- **Theory:** Satellite communication enables global wireless communication by using satellites in orbit. It is used for applications like television broadcasting, internet services, and military communication. Since satellites are in geostationary orbit, they provide global coverage, making them ideal for remote or rural areas where other forms of communication are unavailable.

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9. Cellular (4G/5G)

- **Theory:** Cellular networks, including 4G and 5G, are used for mobile communication and data transfer. 4G provides high-speed internet access, while 5G offers even faster speeds and lower latency. 5G networks are designed to support applications like smart cities, autonomous vehicles, and industrial automation by providing ultra-reliable, low-latency communication.

Summary Table

Technology	Range of Communication	Cost	Propagation Delay	Power Consumption	Throughput
Bluetooth	10-100 meters	₹100-500	1-2 ms	Low	1-3 Mbps
Wi-Fi	30-100 meters (indoor)	₹500-₹2500	1-5 ms	Moderate	1-10 Gbps
Zigbee	100 meters	₹200-1500	<20 ms	Very Low	250 kbps
LoRa	15-20 km	₹500-₹4000	100 ms	Very Low	0.3-50 kbps
5G	Several kilometers	₹15000-₹25000	1 ms	Moderate-High	100 Mbps - 10 Gbps
NFC	10 cm	₹50-₹500	Low	Very Low	106 to 424 kbit/s
Infrared	1-5 meters	₹50-₹100	1 ms	Low	1 Mbps
Satellite	Global	₹50000-₹100000	250-500 ms	High	Varies (kbps to Gbps)
Cellular (4G/5G)	Few kilometers	₹500-₹2000	20-50 ms (4G), 1-10 ms (5G)	Moderate	1 Gbps (4G), 10 Gbps (5G)

Conclusion

Each wireless technology is tailored for specific applications, ranging from short-range, low-power options like Bluetooth and Zigbee to long-range, high-capacity networks like 5G and satellite communication. Understanding these differences helps in choosing the right technology for various devices and use cases, particularly in the IoT and modern communication landscape.